

Why learn how to build recommendation engines?

BUILDING RECOMMENDATION ENGINES WITH PYSPARK

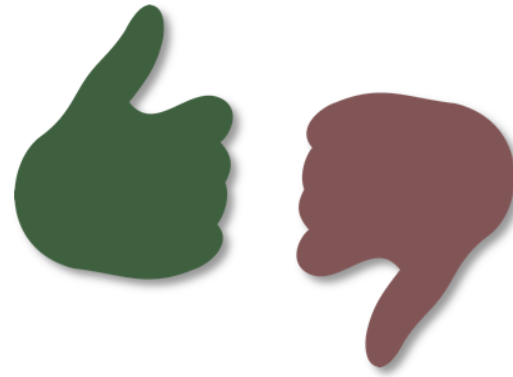
Jamen Long
Data Scientist at Nike



What recommendations look like



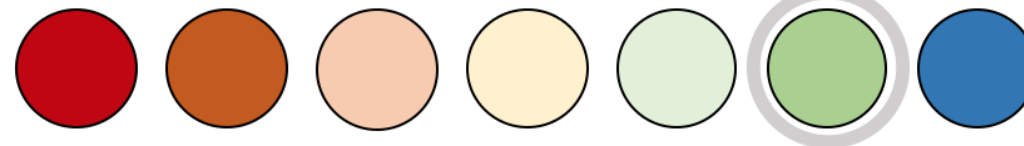
Learning about you



You rated this item:



Worst



Best

How recommendation engines work



Movie	Rating
Dark Knight	5
The Notebook	-
Iron Man	4
Finding Nemo	3
Shrek	-
Tangled	1
Jurassic Park	4

How recommendation engines work



Movie	Rating
Dark Knight	5
The Notebook	-
Iron Man	4
Finding Nemo	3
Shrek	-
Tangled	1
Jurassic Park	4

Movie	Rating
Dark Knight	5
The Notebook	-
Iron Man	5
Finding Nemo	3
Shrek	4
Tangled	2
Jurassic Park	-



How recommendation engines work



Movie	Rating		Movie	Rating
Dark Knight	5	Similar	Dark Knight	5
The Notebook	-		The Notebook	-
Iron Man	4	Similar	Iron Man	5
Finding Nemo	3	Similar	Finding Nemo	3
Shrek	-		Shrek	4
Tangled	1	Similar	Tangled	2
Jurassic Park	4		Jurassic Park	-



How recommendation engines work



Movie	Rating		Movie	Rating
Dark Knight	5	Similar	Dark Knight	5
The Notebook	-		The Notebook	-
Iron Man	4	Similar	Iron Man	5
Finding Nemo	3	Similar	Finding Nemo	3
Shrek	-		Shrek	4.5
Tangled	1	Similar	Tangled	2
Jurassic Park	4		Jurassic Park	-



How recommendation engines work



Movie	Rating		Movie	Rating
Dark Knight	5	Similar	Dark Knight	5
The Notebook	-		The Notebook	-
Iron Man	4	Similar	Iron Man	5
Finding Nemo	3	Similar	Finding Nemo	3
Shrek	-	Recommendation	Shrek	4.5
Tangled	1	Similar	Tangled	2
Jurassic Park	4		Jurassic Park	-



The Power of Recommendation Engines

powered by increasingly sophisticated models that analyze transaction data and digital footprints (for example, what topics are hot on social networks). **Already, 35 percent of what consumers purchase on Amazon and 75 percent of what they watch on Netflix come from product recommendations based on such algorithms.** Company-directed marketing is also competing for attention. Recommendations through social networks, user-generated content, and targeted advertising for the average

Ian Mackenzie, Chris Meyer, and Steve Noble
McKinsey & Company, October 2013

Prerequisites

- [Introduction to PySpark](#)
- [Intermediate Python](#)
- [Supervised Learning with scikit-learn](#)

Let's practice!

BUILDING RECOMMENDATION ENGINES WITH PYSPARK

Recommendation engine types and data types

BUILDING RECOMMENDATION ENGINES WITH PYSPARK



Jamen Long
Data Scientist at Nike

Two types of recommendation engines:

CONTENT-BASED FILTERING

Based on features of items

- Genre: Comedy, Action, Drama
- Animation: Animated, Not animated
- Language: English, Spanish, Korean
- Decade Produced: 1950's, 1980's
- Actors: Meryl Streep, Tom Hanks

COLLABORATIVE FILTERING

Two types of recommendation engines

CONTENT-BASED FILTERING

Based on features of items

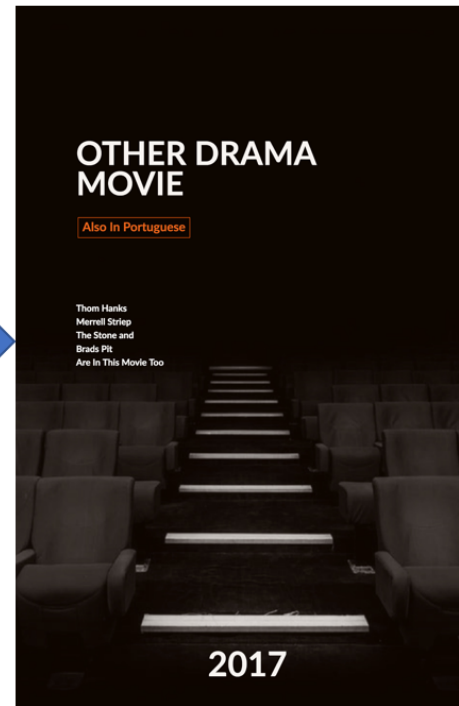
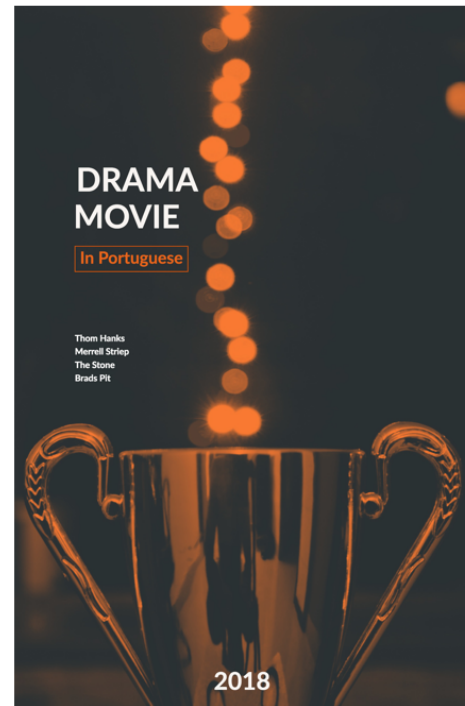


COLLABORATIVE FILTERING

Two types of recommendation engines

CONTENT-BASED FILTERING

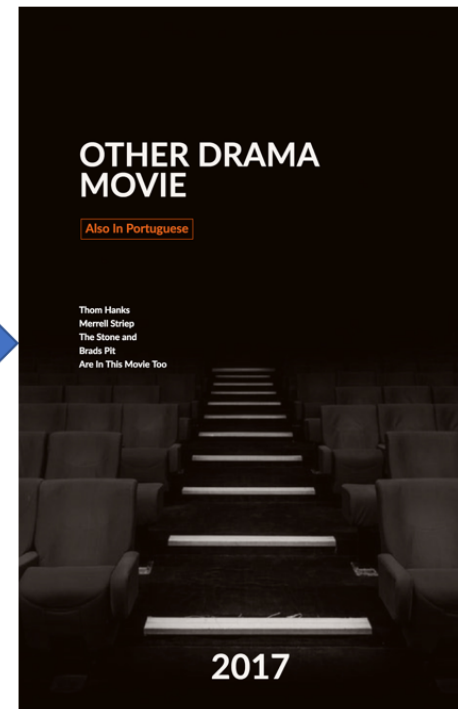
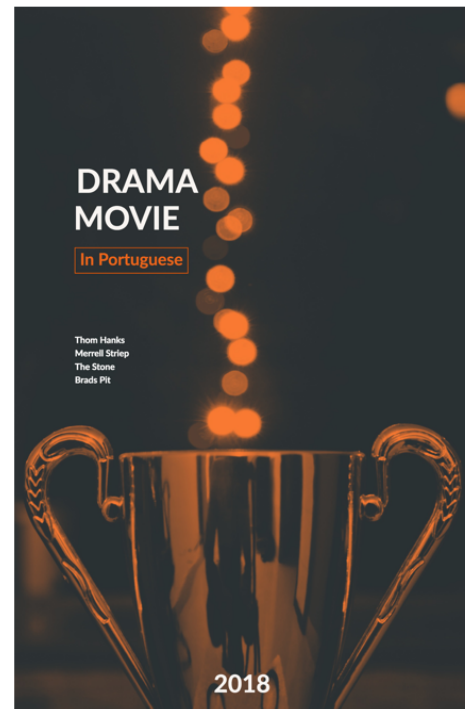
Based on features of items



Two types of recommendation engines

CONTENT-BASED FILTERING

Based on features of items



Recommended for you!

Two types of recommendation engines

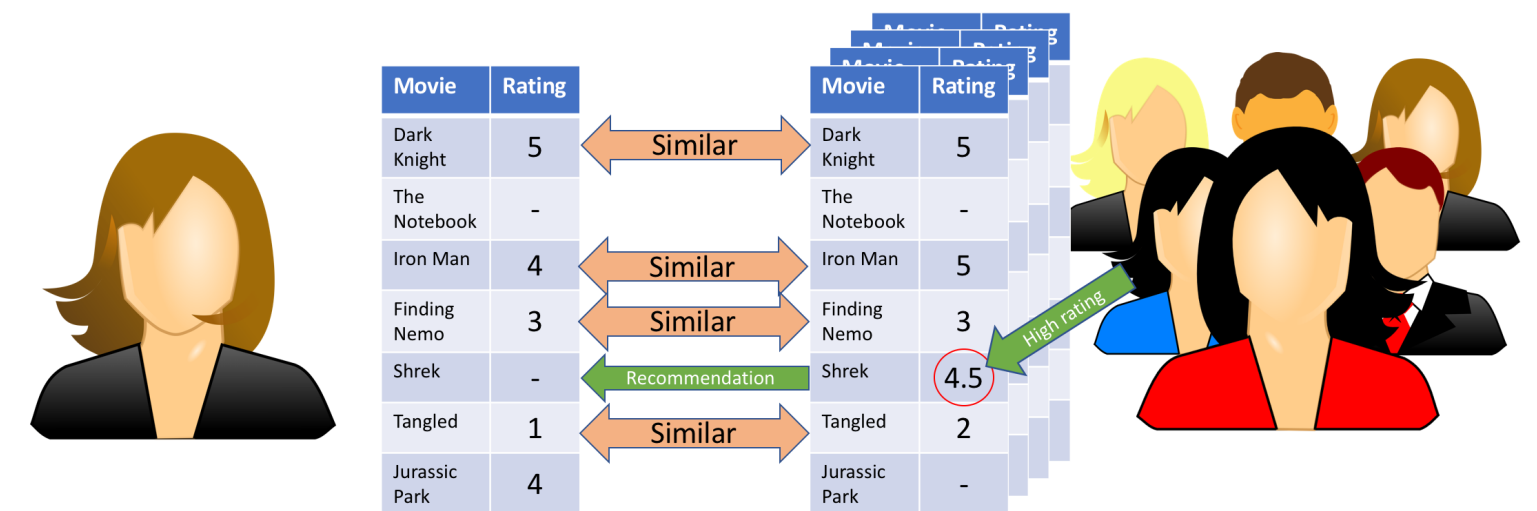
CONTENT-BASED FILTERING

Based on features of items



COLLABORATIVE FILTERING

Based on similar user preferences



Two types of ratings

Two types of ratings

EXPLICIT RATINGS

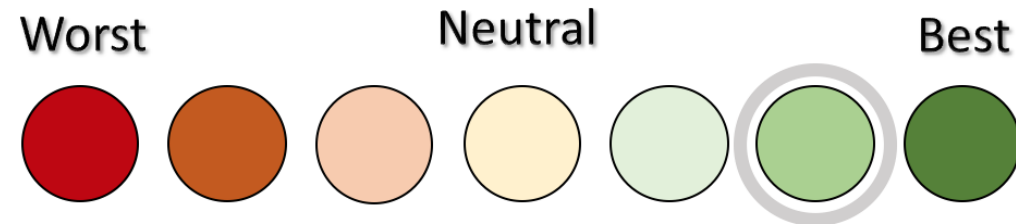
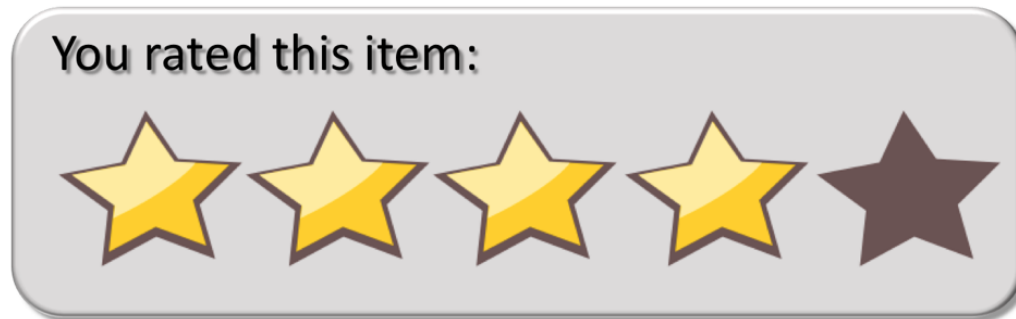
Two types of ratings

EXPLICIT RATINGS

IMPLICIT RATINGS

Two types of ratings

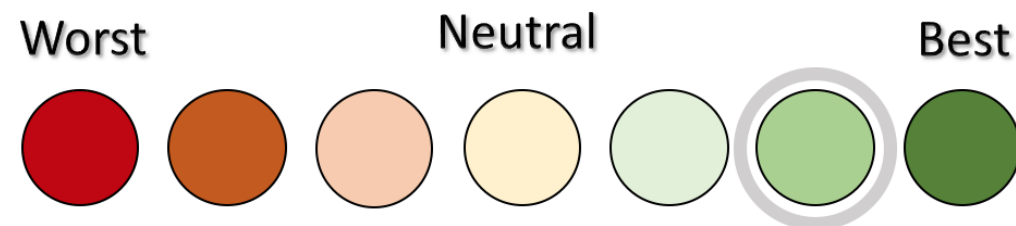
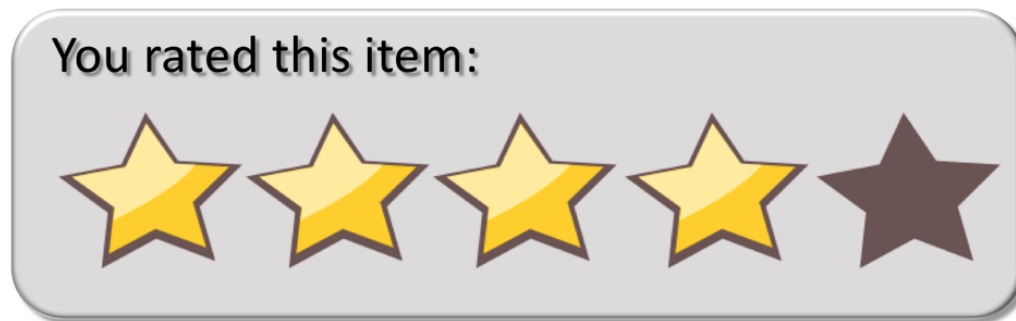
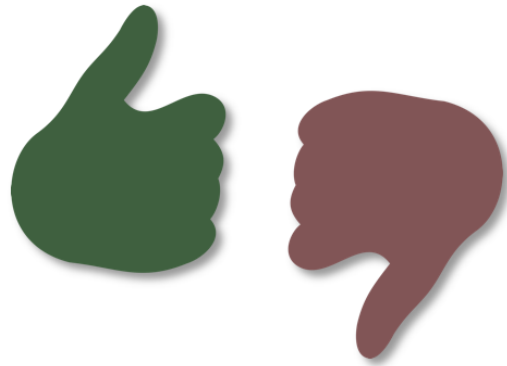
EXPLICIT RATINGS



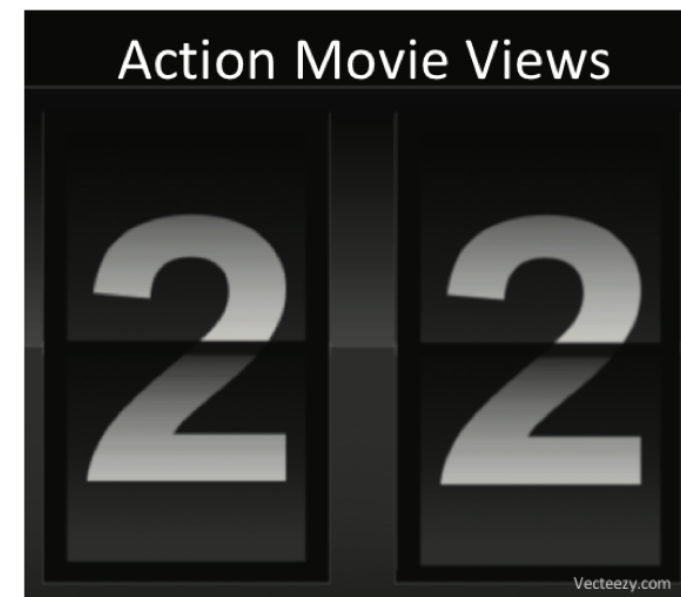
IMPLICIT RATINGS

Two types of ratings

EXPLICIT RATINGS

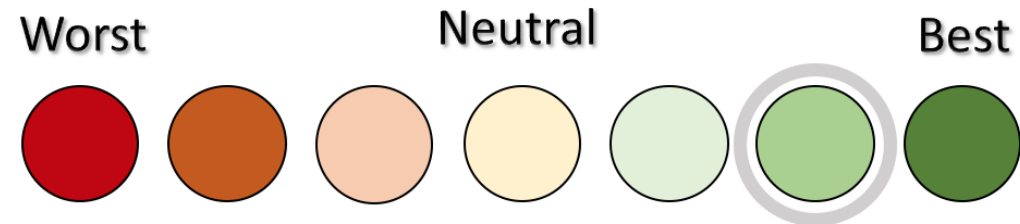
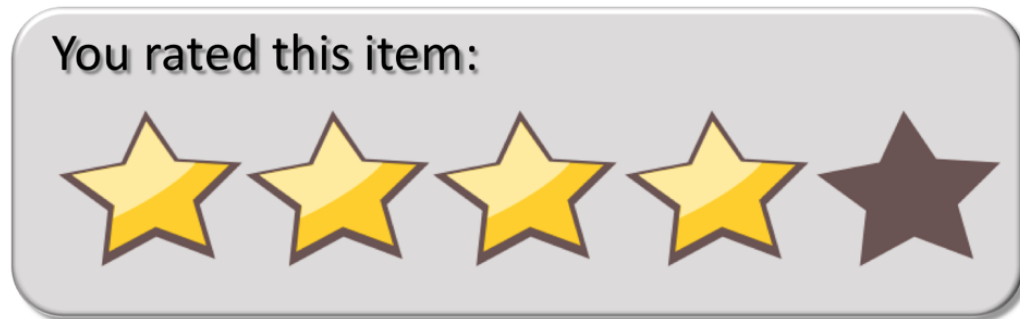


IMPLICIT RATINGS

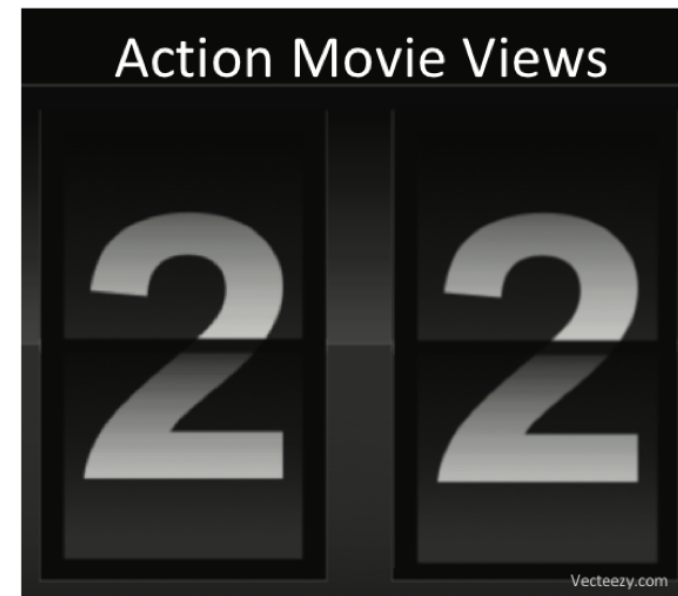


Two types of ratings

EXPLICIT RATINGS



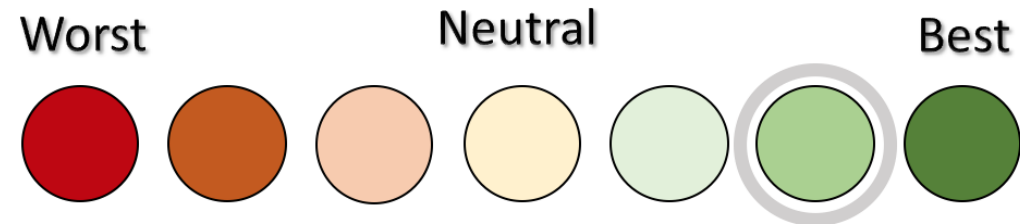
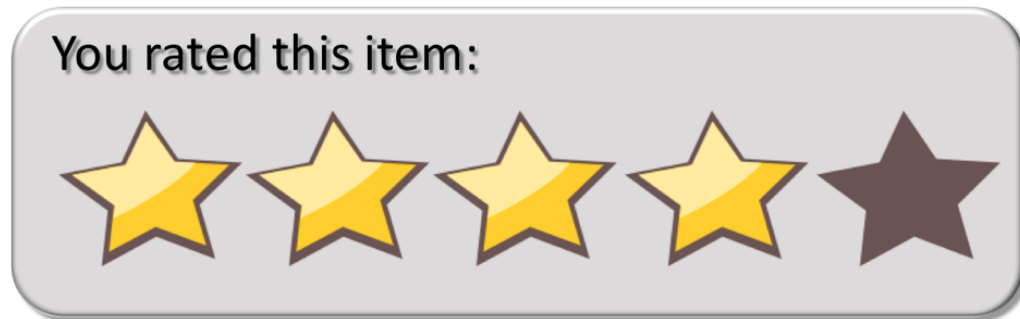
IMPLICIT RATINGS



≈ Low Confidence Rating

Two types of ratings

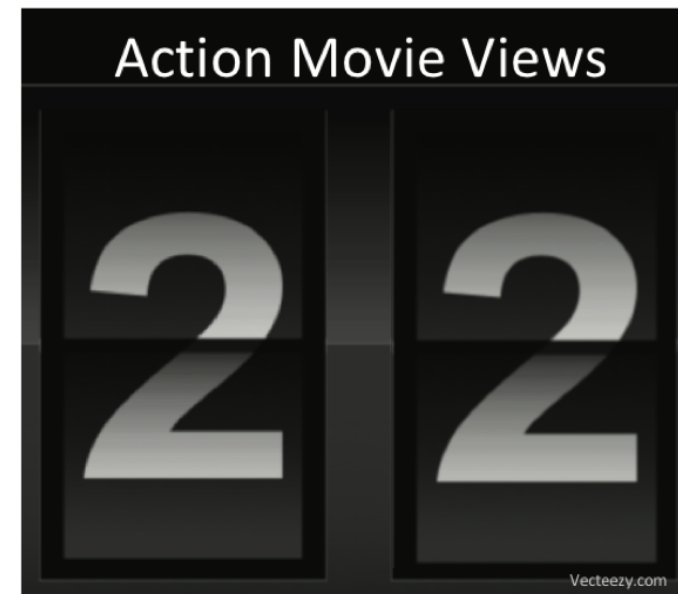
EXPLICIT RATINGS



IMPLICIT RATINGS



≈ Low Confidence Rating



≈ High Confidence Rating

Let's practice!

BUILDING RECOMMENDATION ENGINES WITH PYSPARK

Uses for recommendation engines

BUILDING RECOMMENDATION ENGINES WITH PYSPARK



Jamen Long
Data Scientist at Nike

Original Ratings Matrix

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie ...	Movie <i>n</i>
User 1							
User 2							
User 3							
User 4							
User 5							
User...							
User <i>m</i>							

Original Ratings Matrix

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie ...	Movie <i>n</i>
User 1							
User 2							
User 3							
User 4							
User 5							
User...							
User <i>m</i>							



X

Factor Matrices

Original Ratings Matrix

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie ...	Movie <i>n</i>
User 1							
User 2							
User 3							
User 4							
User 5							
User...							
User <i>m</i>							



	LF1	LF...	LFk
User 1			
User 2			
User 3			
User 4			
User 5			
User...			
User <i>m</i>			

X

Factor Matrices

Original Ratings Matrix

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie ...	Movie <i>n</i>
User 1							
User 2							
User 3							
User 4							
User 5							
User...							
User <i>m</i>							



	LF1	LF...	LFk
User 1			
User 2			
User 3			
User 4			
User 5			
User...			
User <i>m</i>			

X

Factor Matrices

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie ...	Movie <i>n</i>
LF 1							
LF...							
LFk							

Original Ratings Matrix

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie ...	Movie <i>n</i>
User 1							
User 2							
User 3							
User 4							
User 5							
User...							
User <i>m</i>							

ALS

	LF1	LF...	LFk
User 1			
User 2			
User 3			
User 4			
User 5			
User...			
User <i>m</i>			

X

Factor Matrices

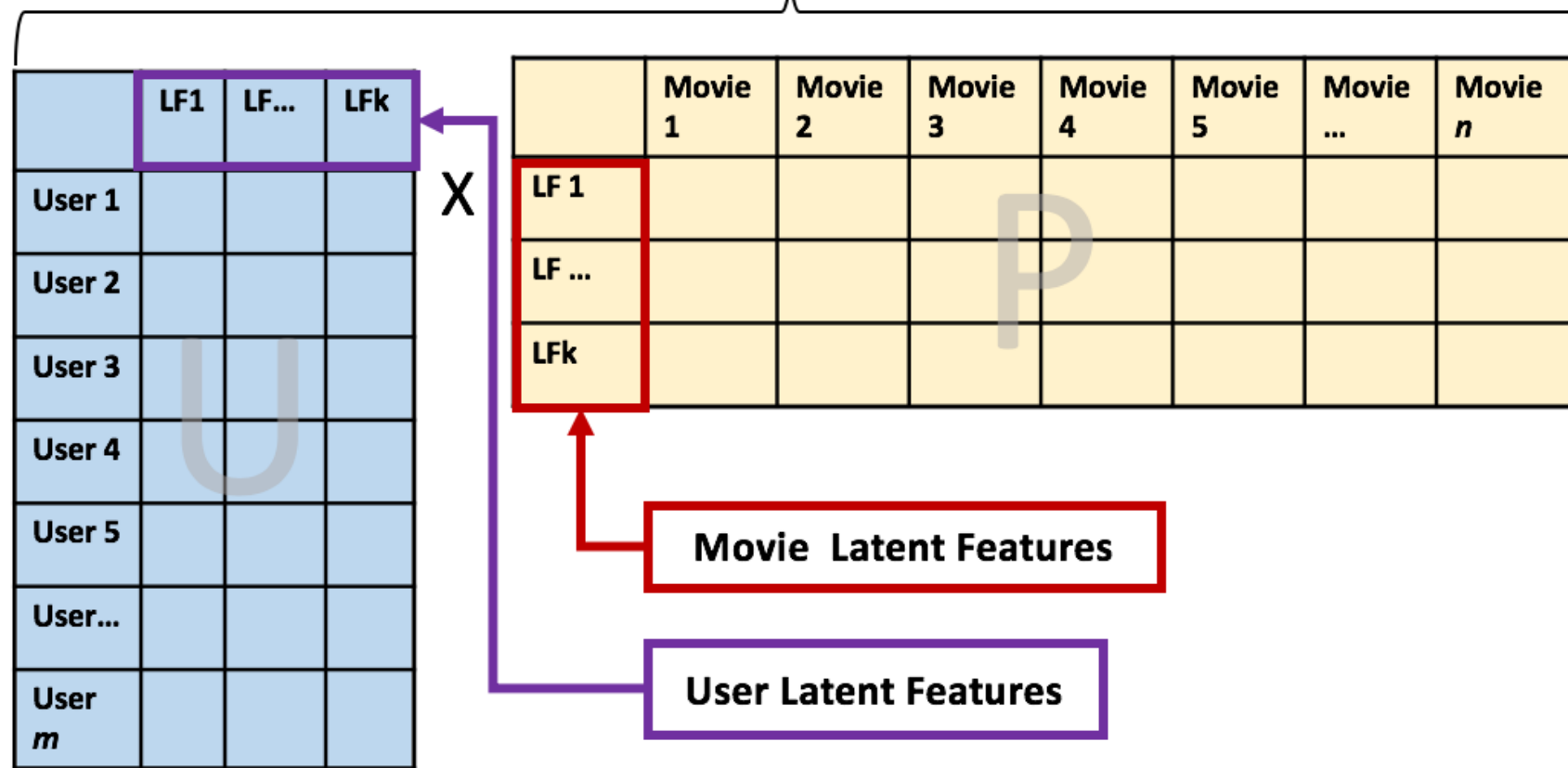
	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie ...	Movie <i>n</i>
LF 1							
LF...							
LFk							

Original Ratings Matrix

	Movie 1	Movie 2	Movie 3	Movie 4	Movie 5	Movie ...	Movie <i>n</i>
User 1							
User 2							
User 3							
User 4							
User 5							
User...							
User <i>m</i>							



Factor Matrices



Original Ratings Matrix

	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Drama MovN
User 1	5	5	1	1	1	...	1
User 2	4	4	1	1	1	...	1
User 3	4	5	1	1	1	...	1
User 4	2	2	1	1	1	...	1
User 5	1	1	1	5	5	...	5
User...	...	...	...	...	...	...	...
User <i>m</i>	1	1	1	4	5	...	5



Factor Matrices

	LF1	LF...	LFk
User 1	2.2	...	.00
User 2	1.9	...	.00
User 3	2.1	...	.01
User 4	.8	...	.01
User 5	.01	...	2.3
User...		...	...
User <i>m</i>	.00	...	2.2

 $\times$

	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Movie <i>n</i>
LF 1	2.1	2.2	.01	.2	.04	...	.1
LF ...	...	...	...	...	...	...	...
LFk	.1	.01	.04	2.0	2.1	...	2.2

Original Ratings Matrix

	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Drama MovN
User 1	5	5	1	1	1	...	1
User 2	4	4	1	1	1	...	1
User 3	4	5	1	1	1	...	1
User 4	2	2	1	1	1	...	1
User 5	1	1	1	5	5	...	5
User...	...	...	...	...	...	...	...
User <i>m</i>	1	1	1	4	5	...	5



Factor Matrices

	LF1	LF...	LFk
User 1	2.2	...	.00
User 2	1.9	...	.00
User 3	2.1	...	.01
User 4	.8	...	.01
User 5	.01	...	2.3
User...	...	...	...
User <i>m</i>	.00	...	2.2

 $\times$

	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Movie <i>n</i>
LF 1	2.1	2.2	.01	.2	.04	...	.1
LF ...	...	...	...	...	...	...	...
LFk	.1	.01	.04	2.0	2.1	...	2.2

Original Ratings Matrix

[No Title]	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Drama MovN
User 1	5	5	1	1	1	...	1
User 2	4	4	1	1	1	...	1
User 3	4	5	1	1	1	...	1
User 4	2	2	1	1	1	...	1
User 5	1	1	1	5	5	...	5
User...	...	...	...	...	...	...	...
User <i>m</i>	1	1	1	4	5	...	5

ALS

	LF1	LF...	LFk
User 1	2.2	...	.00
User 2	1.9	...	.00
User 3	2.1	...	.01
User 4	.8	...	.01
User 5	.01	...	2.3
User...		...	...
User <i>m</i>	.00	...	2.2

 $\times$

	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Movie <i>n</i>
LF 1	2.1	2.2	.01	.2	.04	...	.1
LF ...	...	...	...	...	...	...	...
LFk	.1	.01	.04	2.0	2.1	...	2.2

Original Ratings Matrix

	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Drama MovN
User 1	5	5	1	1	1	...	1
User 2	4	4	1	1	1	...	1
User 3	4	5	1	1	1	...	1
User 4	2	2	1	1	1	...	1
User 5	1	1	1	5	5	...	5
User...	...	...	...	...	...	...	...
User <i>m</i>	1	1	1	4	5	...	5



Factor Matrices

	LF1	LF...	LFk
User 1	2.2	...	.00
User 2	1.9	...	.00
User 3	2.1	...	.01
User 4	.8	...	.01
User 5	.01	...	2.3
User...		...	...
User <i>m</i>	.00	...	2.2

 $\times$

	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Movie <i>n</i>
LF 1	2.1	2.2	.01	.2	.04	...	.1
LF ...	...	...	...	...	...	...	...
LFk	.1	.01	.04	2.0	2.1	...	2.2

Original Ratings Matrix

	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Drama MovN
User 1	5	5	1	1	1	...	1
User 2	4	4	1	1	1	...	1
User 3	4	5	1	1	1	...	1
User 4	2	2	1	1	1	...	1
User 5	1	1	1	5	5	...	5
User...	...	...	...	...	...	...	...
User <i>m</i>	1	1	1	4	5	...	5



Factor Matrices

	LF1	LF...	LFk
User 1	2.2	...	.00
User 2	1.9	...	.00
User 3	2.1	...	.01
User 4	.8	...	.01
User 5	.01	...	2.3
User...	...	...	...
User <i>m</i>	.00	...	2.2

 $\times$

	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Movie <i>n</i>
LF 1	2.1	2.2	.01	.2	.04	...	.1
LF ...	...	...	...	...	...	...	...
LFk	.1	.01	.04	2.0	2.1	...	2.2

Original Ratings Matrix

	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Drama MovN
User 1	5	5	1	1	1	...	1
User 2	4	4	1	1	1	...	1
User 3	4	5	1	1	1	...	1
User 4	2	2	1	1	1	...	1
User 5	1	1	1	5	5	...	5
User...	...	...	...	...	...	...	...
User <i>m</i>	1	1	1	4	5	...	5

ALS

Factor Matrices

	LF1	LF...	LFk
User 1	2.2	...	.00
User 2	1.9	...	.00
User 3	2.1	...	.01
User 4	.8	...	.01
User 5	.01	...	2.3
User...	...	...	...
User <i>m</i>	.00	...	2.2

 $\times$

	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Movie <i>n</i>
LF 1	2.1	2.2	.01	.2	.04	...	.1
LF ...	...	...	...	...	...	...	...
LFk	.1	.01	.04	2.0	2.1	...	2.2

Horror Movies Scored Low

Dramas Scored High

Original Ratings Matrix

	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Drama MovN
User 1	5	5	1	1	1	...	1
User 2	4	4	1	1	1	...	1
User 3	4	5	1	1	1	...	1
User 4	2	2	1	1	1	...	1
User 5	1	1	1	5	5	...	5
User...	...	...	...	...	...	...	...
User <i>m</i>	1	1	1	4	5	...	5

ALS

	LF1	LF...	LFk
User 1	2.2	...	.00
User 2	1.9	...	.00
User 3	2.1	...	.01
User 4	.8	...	.01
User 5	.01	...	2.3
User...		...	...
User <i>m</i>	.00	...	2.2

X

	Horror Movies Scored High			Dramas Scored Low			
	Horror Mov1	Horror Mov2	Horror Mov3	Drama Mov1	Drama Mov2	Movie ...	Drama MovN
LF 1	2.1	2.2	.01	.2	.04	...	.1
LF ...	...	...	...	...	...	...	...
LFk	.1	.01	.04	2.0	2.1	...	2.2
	Horror Movies Scored Low			Dramas Scored High			

Original Matrix

	Movie 1	Movie 2	Movie3	Movie...	Movie <i>n</i>
User 1	5	4.8	4.9	...	4.9
User 2	4.5	4.3	4.4	...	4.3
User 3	4.8	4.6	4.7	...	4.6
User 4	2.1	1.98	2	...	2
User 5	4.1	4	4	...	4
User...	...	...	...	...	...
User <i>m</i>	4.6	4.4	4.5	...	4.4

ALS

	LF 1	LF...	LF <i>k</i>
User 1	2.2	...	0.0
User 2	1.95	...	0.0
User 3	2.1	...	0.01
User 4	0.90	...	0.01
User 5	1.8	...	0.03
User...	...	...	...
User <i>m</i>	2.0	...	0.02

X

	Movie 1	Movie 2	Movie3	Movie...	Movie <i>n</i>
LF 1	2.3	2.2	2.23	...	2.21
LF...	...	...	...	...	...
LF <i>k</i>	0.1	0.01	0.02	...	0.02

Factor Matrices

Original Matrix

	Movie 1	Movie 2	Movie3	Movie...	Movie <i>n</i>
User 1	5	4.8	4.9	...	4.9
User 2	4.5	4.3	4.4	...	4.3
User 3	4.8	4.6	4.7	...	4.6
User 4	2.1	1.98	2	...	2
User 5	4.1	4	4	...	4
User...	...	...	...	...	...
User <i>m</i>	4.6	4.4	4.5	...	4.4

ALS

	LF 1	LF...	LF <i>k</i>
User 1	2.2	...	0.0
User 2	1.95	...	0.0
User 3	2.1	...	0.01
User 4	0.90	...	0.01
User 5	1.8	...	0.03
User...	...	...	...
User <i>m</i>	2.0	...	0.02

X

Factor Matrices

	Movie 1	Movie 2	Movie3	Movie...	Movie <i>n</i>
LF 1	2.3	2.2	2.23	...	2.21
LF...	...	...	...	...	...
LF <i>k</i>	0.1	0.01	0.02	...	0.02

Original Matrix

	Movie 1	Movie 2	Movie3	Movie...	Movie <i>n</i>
User 1	5	4.8	4.9	...	4.9
User 2	4.5	4.3	4.4	...	4.3
User 3	4.8	4.6	4.7	...	4.6
User 4	2.1	1.98	2	...	2
User 5	4.1	4	4	...	4
User...	...	...	...	...	...
User <i>m</i>	4.6	4.4	4.5	...	4.4

ALS

	LF 1	LF...	LF <i>k</i>
User 1	2.2	...	0.0
User 2	1.95	...	0.0
User 3	2.1	...	0.01
User 4	0.90	...	0.01
User 5	1.8	...	0.03
User...	...	...	...
User <i>m</i>	2.0	...	0.02

X

Factor Matrices

	Movie 1	Movie 2	Movie3	Movie...	Movie <i>n</i>
LF 1	2.3	2.2	2.23	...	2.21
LF...	...	...	...	...	...
LF <i>k</i>	0.1	0.01	0.02	...	0.02

Movie 1: "The Lion King"

Movie 2: "10 Things I Hate About You"

Movie 4: "West Side Story"

Movie *n*: "She's the Man"

Original Matrix

	Movie 1	Movie 2	Movie3	Movie...	Movie <i>n</i>
User 1	5	4.8	4.9	...	4.9
User 2	4.5	4.3	4.4	...	4.3
User 3	4.8	4.6	4.7	...	4.6
User 4	2.1	1.98	2	...	2
User 5	4.1	4	4	...	4
User...	...	...	...	...	...
User <i>m</i>	4.6	4.4	4.5	...	4.4
<i>Avg Rating</i>	4.2	4.0	4.1	...	4

ALS

Factor Matrices

	LF 1	LF...	LF k
User 1	2.2	...	0.0
User 2	1.95	...	0.0
User 3	2.1	...	0.01
User 4	0.90	...	0.01
User 5	1.8	...	0.03
User...	...	...	...
User <i>m</i>	2.0	...	0.02

X

	Movie 1	Movie 2	Movie3	Movie...	Movie <i>n</i>
LF 1	2.3	2.2	2.23	...	2.21
LF...	...	...	...	...	...
LF k	0.1	0.01	0.02	...	0.02

Movie 1: "The Lion King"

Movie 2: "10 Things I Hate About You"

Movie 4: "West Side Story"

Movie *n*: "She's the Man"

Movie 1: "The Lion King"
 Movie 2: "10 Things I Hate About You"
 Movie 4: "West Side Story"
 Movie *n*: "She's the Man"

Original Matrix

	Movie 1	Movie 2	Movie3	Movie...	Movie n
User 1	5	4.8	4.9	...	4.9
User 2	4.5	4.3	4.4	...	4.3
User 3	4.8	4.6	4.7	...	4.6
User 4	2.1	1.98	2	...	2
User 5	4.1	4	4	...	4
User...	...	...	...	...	...
User m	4.6	4.4	4.5	...	4.4
<i>Avg Rating</i>	4.2	4.0	4.1	...	4

ALS

	LF 1	LF...	LF k
User 1	2.2	...	0.0
User 2	1.95	...	0.0
User 3	2.1	...	0.01
User 4	0.90	...	0.01
User 5	1.8	...	0.03
User...	...	...	...
User m	2.0	...	0.02

X

	Movie 1	Movie 2	Movie3	Movie...	Movie n
LF 1	2.3	2.2	2.23	...	2.21
LF...	...	...	...	...	...
LF k	0.1	0.01	0.02	...	0.02

Movie 1: "The Lion King" ("Hamlet")

Movie 2: "10 Things I Hate About You" ("Taming of the Shrew")

Movie 4: "West Side Story" ("Romeo and Juliet")

Movie n : "She's the Man" ("Twelfth Night")

Let's practice!

BUILDING RECOMMENDATION ENGINES WITH PYSPARK